

Comparative Analysis of YOLO Models for Object Detection

Table 1: Comparison of YOLO Models Based on Training Metrics

Model	Epochs	Train Box Loss (Start → End)	Train Cls Loss (Start → End)	Train Dfl Loss (Start → End)	Val Box Loss (Start → End)	Val Cls Loss (Start → End)	Val Dfl Loss (S
Inferred v9	40	3.6 → 2.6	5.0 → 2.0	3.5 → 1.5	3.5 → 2.6	3.5 → 1.5	3.0 →
Inferred v10	40	1.7 → 1.2	2.5 → 1.0	1.25 → 1.05	1.7 → 1.3	2.5 → 1.0	1.35 →
Model 5	40	1.6 → 1.2	2.0 → 0.75	1.2 → 1.05	1.6 → 1.2	1.6 → 0.6	1.2 →
YOLO v5	40	0.10 → 0.04	0.030 → 0.005	N/A (Obj: 0.050 → 0.025)	0.08 → 0.04	0.015 → 0.0025	N/A (Obj: 0.0
YOLOv7	40	0.07 → 0.03	0.0125 → 0.0025	N/A (Obj: 0.016 → 0.010)	0.10 → 0.05	0.030 → 0.005	N/A (Obj: 0.0
YOLOv9	40	1.7 → 1.2	2.5 → 1.0	1.25 → 1.05	1.7 → 1.3	3.0 → 1.0	1.35 →
YOLOv11	100	1.6 → 1.1	1.6 → 0.6	1.2 → 0.95	1.6 → 1.2	1.4 → 0.6	1.25 →
YOLOv12	100	1.6 → 1.1	1.6 → 0.6	1.2 → 0.95	1.6 → 1.2	1.4 → 0.6	1.25 →

Conclusion

The comparative analysis of the YOLO models reveals that YOLOv12 (also identified as YOLOv11 in some contexts) is the top performer. It achieves the highest mAP50 (~ 0.8 , tied with YOLO v5 and YOLOv7) and mAP50-95 (~ 0.5), demonstrating superior detection accuracy and robustness across IoU thresholds. Additionally, YOLOv12 has the highest recall (~ 0.7) among the models, indicating better detection of true positives, and its precision (~ 0.9) matches the best performers. The extended training over 100 epochs contributes to its low losses and high metrics, making it an ideal choice for applications requiring precise object detection, such as autonomous navigation. For scenarios prioritizing efficiency, YOLO v5 and YOLOv7 are strong alternatives due to their low losses and high mAP50 over fewer epochs.